

Faculty of Engineering & Technology

Department of Electrical & Computer Engineering

Real-Time and Interfacing Techniques Laboratory

Project – Smart Egg Incubator

ENCS5140

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Section: 1

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1. Introduction

This report documents the final implementation and validation of a Smart Egg Incubator based on an ESP32 running FreeRTOS. The prototype performs local temperature and humidity monitoring, relay-based actuator control, automatic egg turning, local OLED status display, audible safety alarms, incubation-day tracking, and remote monitoring/manual control through the Blynk IoT platform.

Development was performed incrementally. Wokwi was used for early functional simulation, while the final software was tested on a physical ESP32 with a DHT11 sensor, SSD1306 OLED, relay modules, 28BYJ-48 stepper motor with ULN2003 driver, and buzzer. The design keeps the control and safety logic local to the ESP32 so loss of Wi-Fi does not stop the incubator control loop.

1.1 Learning Objectives Addressed

- Decompose the application into independent FreeRTOS tasks with explicit periods and priorities.
- Use a queue and mutex-protected shared state for communication between concurrent tasks.
- Implement hysteresis-based temperature and humidity control with fail-safe relay states.
- Integrate local sensing, actuation, display, alarms, and egg-turning with an ESP32.
- Integrate Blynk telemetry and remote test controls without making local control dependent on the cloud.
- Detect sensor faults and persistent out-of-range conditions and respond safely.

2. System Architecture

The implemented system follows three layers: sensing, local real-time control, and outputs/remote monitoring. The DHT11 publishes the latest temperature/humidity sample to a one-element FreeRTOS queue. Control, safety, display, and Blynk tasks read that latest sample. Shared operating state is protected primarily by a state mutex. All heater, humidifier, turning, alarm, and day decisions are made locally on the ESP32.

Layer	Implemented Components
Sensing / Input	DHT11 temperature and humidity sensor on GPIO4; Blynk manual/test commands.
ESP32 + FreeRTOS	SensorTask, SafetyTask, HeaterTask, HumidityTask, EggTurnTask, DayTask, AlarmTask, DisplayTask, and BlynkTask.
Actuation / Output	Heater relay on GPIO16, humidifier relay on GPIO17, 28BYJ-48 + ULN2003 on GPIO18/27/25/26, passive buzzer on GPIO19, SSD1306 OLED on GPIO21/22.
Remote Layer	Blynk dashboard for live telemetry, state indicators, manual/test mode, manual turn request, and alarm event transmission.

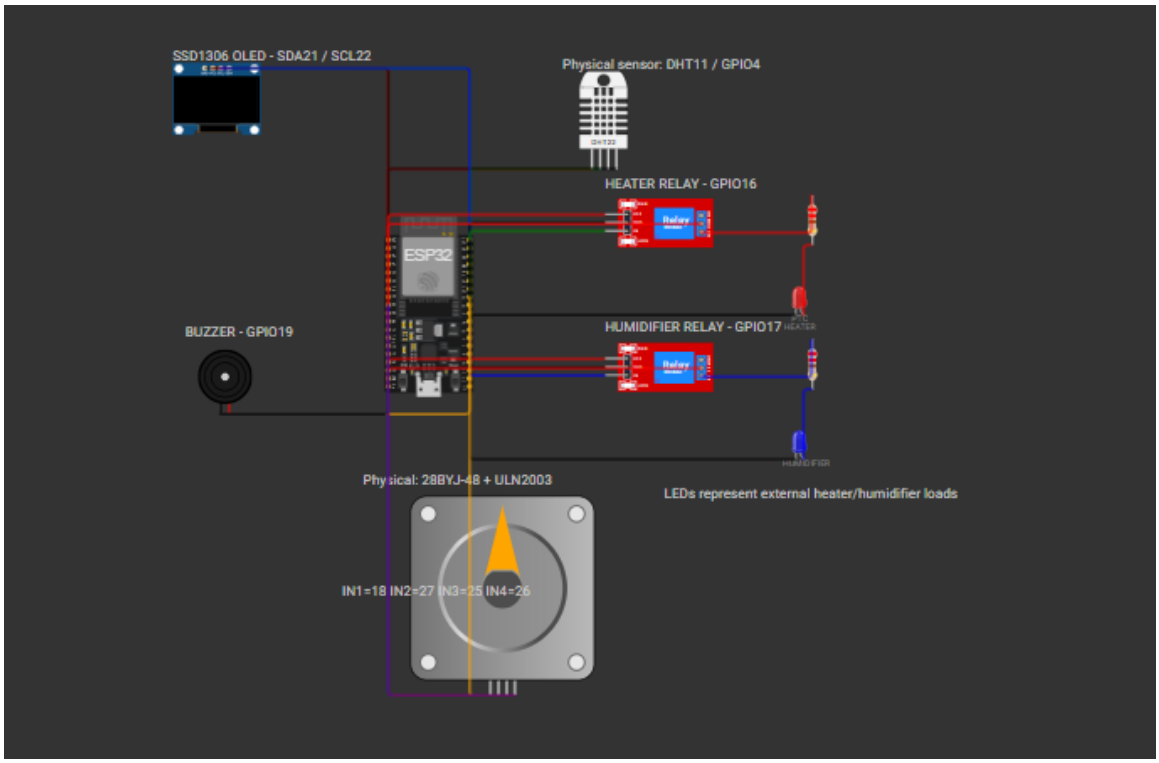


Figure 1. Final connection diagram used for the report. LEDs in the Wokwi drawing represent the external heater/humidifier loads; the physical turning actuator is a 28BYJ-48 with ULN2003.

3. Requirements Coverage

3.1 Functional Requirements

ID	Requirement	Status
FR1	Measure temperature and humidity at least once per second.	Met - SensorTask samples every 1000 ms.
FR2	Maintain 37.2-37.8 C using a relay/SSR-controlled heater.	Met in control logic and relay switching tests. Heater turns ON below 37.2 C and OFF above 37.8 C. Full enclosure thermal calibration remains a bench limitation.
FR3	Maintain 50-55% RH on days 1-17 and 65-70% in lockdown.	Met in control logic. Day-18 target change was tested. Physical fogger/humidifier was not installed in the final bench prototype; relay output was verified.
FR4	Turn eggs at least 4 times/day and stop at lockdown.	Met - real mode uses 6-hour spacing (4/day); bench mode uses 30 s for demonstration. Automatic turning is blocked from Day 18.
FR5	Track elapsed days and days remaining.	Partially met - day and days remaining are shown locally; day is published to Blynk. Timing uses millis() and is not retained across power loss.

FR6	Publish live telemetry/state/day to Blynk every <=2 s.	Met - temperature, humidity, heater, humidifier, motor, day and fault/manual states are published; telemetry update interval is 2 s.
FR7	Blynk notification + local alarm after configurable grace period.	Met - 120 s grace period; rangeAlarmActive drives the local buzzer and Blynk.logEvent() sends the cloud event.
FR8	Continue local control/display/alarm if Wi-Fi/Blynk is lost.	Met in short disconnect testing - local sensing/control continued and Blynk reconnected when Wi-Fi returned.
FR9	Manual/test mode for heater, humidifier and turning motor.	Met - V6 enables manual mode, V7/V8 control heater/humidifier, and V9 requests an egg turn.
FR10	Treat missing/implausible sensor data as a fault.	Met - invalid/NaN samples immediately force outputs OFF; persistent invalid data for about 2.5 s sets systemFault.

3.2 Non-Functional and Real-Time Requirements

ID	Requirement	Status
NFR1	Sensor sampling period 1 s +/- 50 ms.	Implemented at 1000 ms. Exact jitter was not instrumented, so the +/-50 ms bound was not formally measured.
NFR2	Safety/watchdog highest priority, period <=500 ms.	Met - SafetyTask priority 4, 500 ms period.
NFR3	Sensor-to-actuation latency <=2 s.	Met by task timing: 1 s sensor sampling plus 250 ms control task periods; no separate oscilloscope timing measurement was performed.
NFR4	Blynk networking runs independently from control tasks.	Met - BlynkTask is separate; control tasks do not wait for Blynk. Shared state is protected by stateMutex.
NFR5	Shared resources synchronized by queue/mutex.	Mostly met - sensor data uses sensorQueue and operating state uses stateMutex. systemFault is also read as volatile inside the stepper loop for immediate abort, so the strict no-unsynchronized-global form is not fully satisfied.
NFR6	Safe local operation for at least 24 h without Wi-Fi.	Partially verified - a short Wi-Fi disconnect/reconnect test passed; a continuous 24 h soak test was not completed.
NFR7	Egg turn <=10 s and mutually exclusive with shared bus/display operations.	Partially met - a 512-step turn completes well below 10 s. An explicit motor/display mutual-exclusion mutex was not implemented because the stepper does not use the OLED I2C bus.
NFR8	Single bad sensor reading must not latch actuators ON.	Met - invalid readings force heater/humidifier OFF immediately; transient invalid readings recover automatically.

4. Final Bill of Materials and Substitutions

Component	Final Build	Notes / Substitution
ESP32 DevKit V1	Used	Main controller; FreeRTOS + Wi-Fi/Blynk.
DHT11 temperature/humidity sensor	Used (substitution)	Physical sensor used in final bench build. Wokwi DHT22 was only a simulation/visual stand-in. DHT11 has lower accuracy/resolution.
SSD1306 0.96-inch I2C OLED	Used	Local display at I2C address 0x3C, SDA21/SCL22.
Two 5 V relay modules	Used	Active-LOW relay modules for heater and humidifier control.
TEC1-12706 Peltier element	Bench heater	Used as a low-voltage heat source during heater testing; requires proper heatsinking and a correctly rated supply for sustained operation.
Ultrasonic fogger / humidifier	Not installed in final bench test	Humidifier relay/control logic was tested; physical fogger remains a hardware limitation.
28BYJ-48 5 V stepper + ULN2003	Used	Egg-turning actuator; GPIO18/27/25/26; 512-step demonstration movement.
Passive buzzer	Used	Local audible alarm on GPIO19.
5 V USB power for ESP32	Used	ESP32/control electronics.
Separate 5 V / 2 A adapter	Temporary bench supply	Used only for short TEC functionality tests; it is not a final full-power supply for a TEC1-12706.
DS3231 RTC	Not installed	Recommended in brief; omission means day timing is not preserved across power loss.
Independent thermal cutoff	Not installed	Required for a final safe heater installation; documented as a limitation before sustained real heating.

5. FreeRTOS Task Design - Actual Implementation

The final code contains nine FreeRTOS tasks. The numeric priorities and periods below are the values actually used in the submitted implementation, not the suggested values from the project skeleton.

Task	Responsibility	Period	Priority	Synchronization
SafetyTask	Fault detection, 2-minute local range alarm, fail-safe output shutdown	500 ms	4 (Highest)	sensorQueue, stateMutex
SensorTask	Read DHT11 and publish latest sample	1000 ms	3	sensorQueue overwrite
HeaterTask	Automatic/manual heater relay control	250 ms	2	sensorQueue, stateMutex
HumidityTask	Automatic/manual humidity relay control; lockdown range	250 ms	2	sensorQueue, stateMutex

AlarmTask	Generate passive-buzzer tone for system fault/range alarm	1 ms active / 100 ms idle	2	stateMutex
EggTurnTask	Manual and automatic stepper turning	200 ms check; 6 h real / 30 s bench	1	stateMutex; immediate systemFault check
DayTask	Increment incubation day	1 s check; day=24 h	1	stateMutex
DisplayTask	Update OLED with mode/day/days-left/sensors/outputs	500 ms	1	sensorQueue, stateMutex
BlynkTask	Wi-Fi/Blynk reconnect, telemetry, controls, events	20 ms loop; 2 s publish	1	sensorQueue, stateMutex

5.1 Inter-Task Communication and Synchronization

- sensorQueue is a one-element latest-value mailbox. SensorTask overwrites it; safety/control/display/Blynk tasks use xQueuePeek().
- stateMutex protects shared mode, actuator-state, day, motor-state, manual-command, pending-button-reset, and range-alarm variables.
- pendingTurnButtonReset lets EggTurnTask request a Blynk V9 UI reset without making a Blynk API call directly from the motor task.
- The automatic turn interval is implemented with millis()-based timing inside EggTurnTask. The final build does not use a FreeRTOS software timer.
- The motor loop checks systemFault during each step so a persistent safety fault can abort a turn immediately.

6. Blynk IoT Dashboard

The Blynk dashboard is implemented as a remote HMI. It is not required for the local control loops to function. The final virtual-pin mapping is listed below.

V-Pin	Widget	Function
V0	Temperature gauge	Live temperature (C)
V1	Humidity gauge	Live relative humidity (%)
V2	LED / state	Heater actual ON/OFF state
V3	LED / state	Humidifier actual ON/OFF state
V4	Value	Incubation day
V5	LED / state	System fault state
V6	Switch	Manual / Automatic mode
V7	Switch	Manual heater test command
V8	Switch	Manual humidifier test command

V9	Button	Turn Eggs Now request; automatically reset after completion
V10	LED	Motor active state, updated immediately during a turn

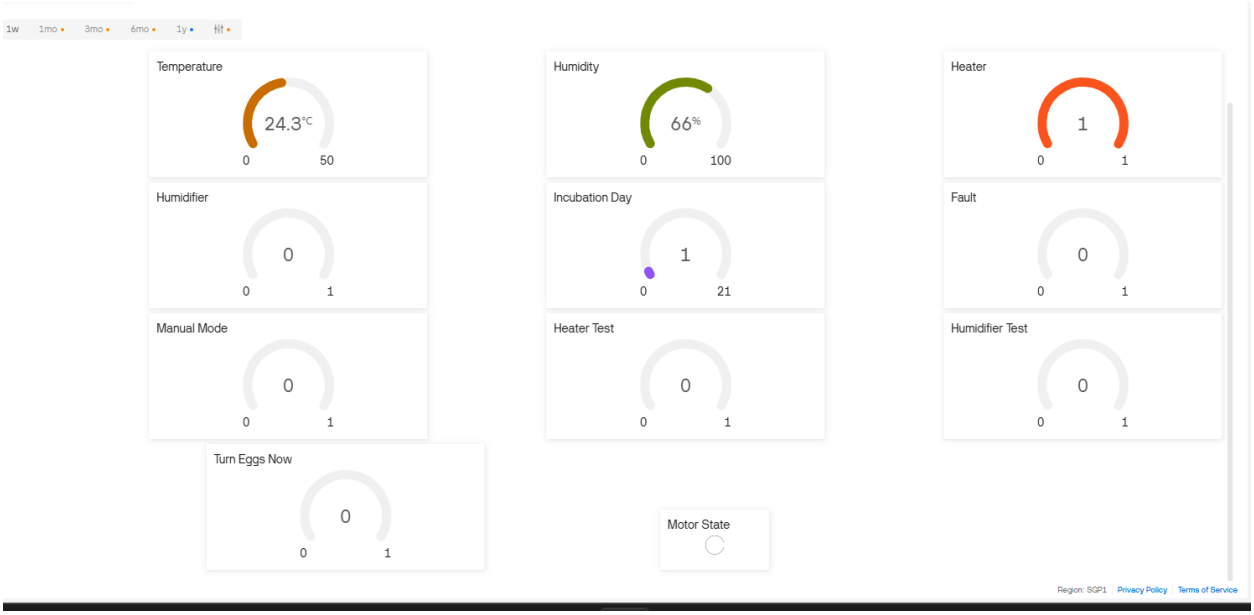
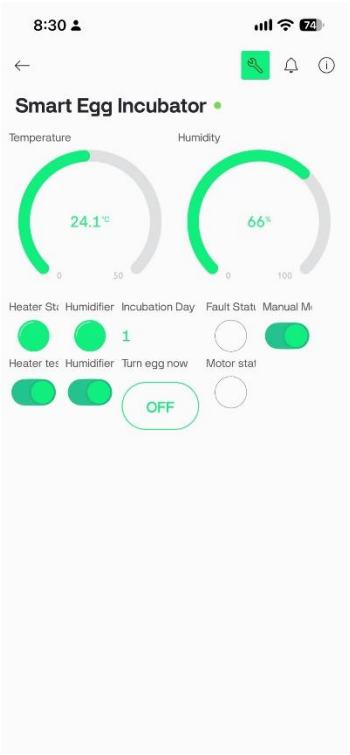
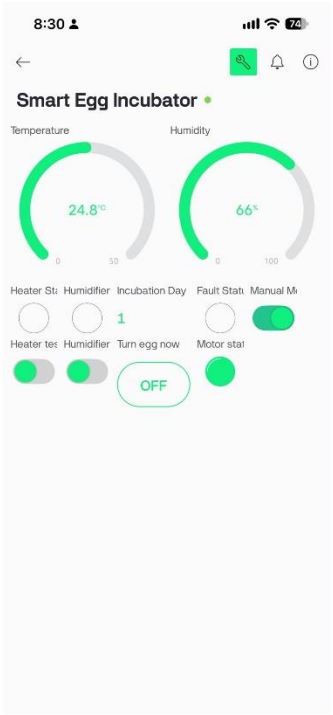


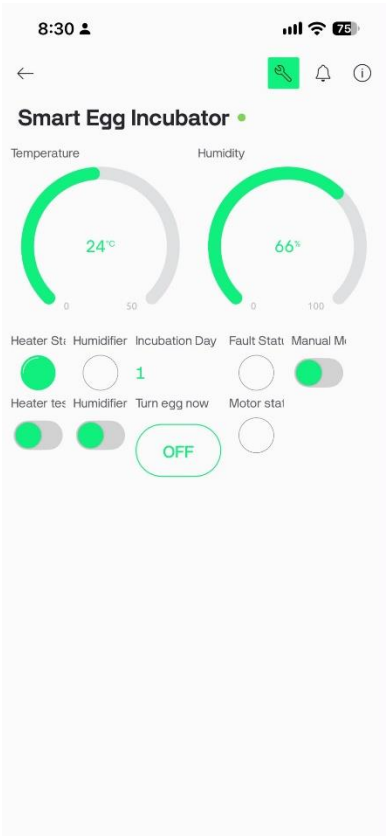
Figure 2. Blynk web dashboard showing live temperature, humidity, heater state, humidifier state, incubation day, fault state, manual controls, and motor state.



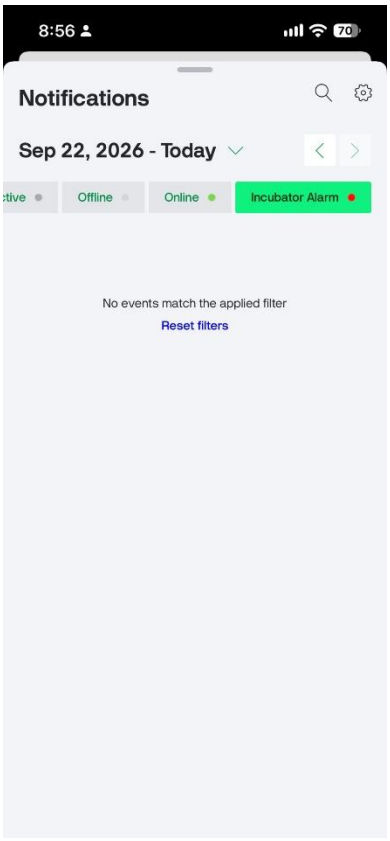
Manual/Test mode



Manual egg-turn; Motor State active



Automatic mode



Incubator Alarm event category configured

Figures 3-5. Blynk mobile evidence: manual actuator control, motor-state indication during a turn, automatic operation, and the configured Incubator Alarm notification category. Event transmission is confirmed by the serial log in Figure 6.

7. Testing and Validation

The system was tested incrementally on the bench. BENCH_TEST_MODE shortened the automatic turn interval to 30 s to make motor behavior observable during the demonstration; real mode uses a 6-hour interval. The following results summarize tests that were actually observed in the serial logs and Blynk dashboard.

Test	Input / Condition	Observed Result	Result
Sensor sampling	Valid DHT11 data at 1 s intervals	Continuous temperature/humidity samples observed.	Pass
Transient sensor error	1-2 invalid samples	Heater/humidifier forced OFF immediately; no system fault if data recovers before ~2.5 s.	Pass
Persistent sensor error	Several consecutive invalid samples	[SAFETY] FAULT DETECTED, then automatic fault clear when valid data returns.	Pass
Automatic heater	Room temperature below 37.2 C	Heater relay commanded ON; turns OFF on invalid/fault.	Pass
Manual controls	Manual mode + V7/V8	Heater and humidifier test commands accepted only in Manual mode.	Pass

Automatic turning	Bench interval = 30 s	Alternating RIGHT/LEFT turn completes; V10 changes ON/OFF.	Pass
Day 18 lockdown	Day temporarily set to 18	Automatic turning stops; humidity band changes to 65-70%. At 64% RH, humidifier output turns ON.	Pass
Local 2-minute alarm	Temperature/humidity kept outside target range	[LOCAL ALARM] ACTIVE after grace period; buzzer remains active until condition clears or Manual/Test mode is selected.	Pass
Blynk event	Same out-of-range test	Serial confirms [ALARM] Blynk notification sent after 2-minute event timer.	Pass
Wi-Fi loss	Wi-Fi disconnected and restored	Local sensing/control continued; Blynk reconnected afterward.	Pass (short test)
Stability	Short unattended integration run	No repeat crash in the later short run; one watchdog timeout had occurred in an earlier integration run.	Partial - longer soak recommended

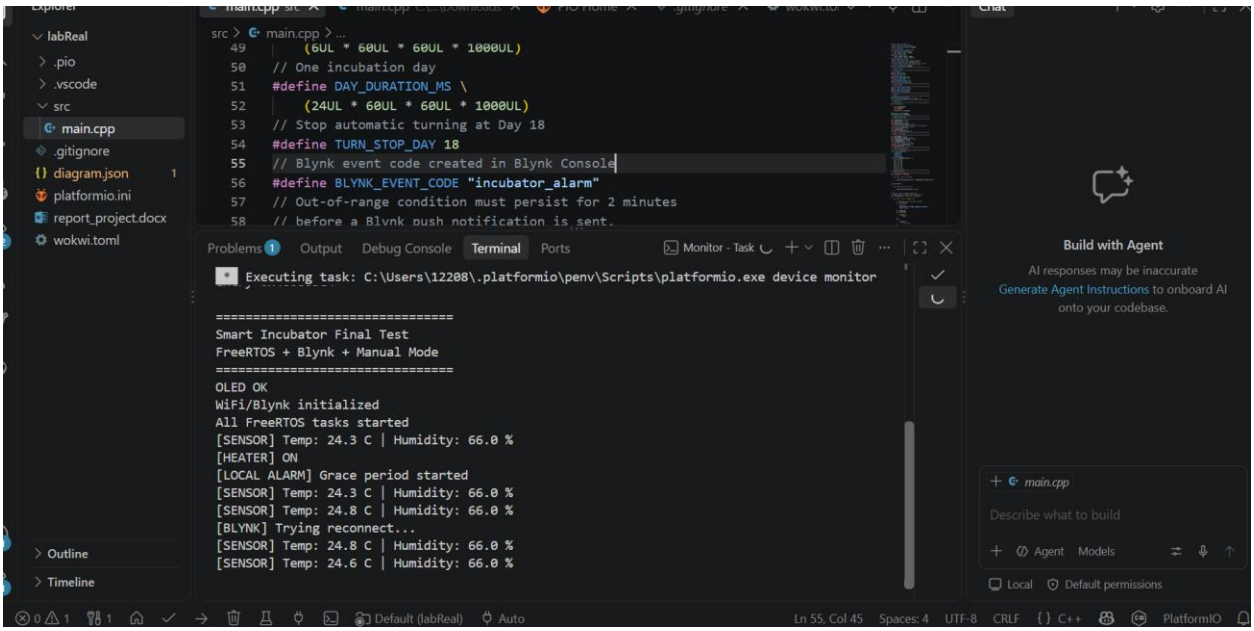


Figure 6. Serial Monitor evidence showing the local alarm active, automatic egg turning, Blynk motor-state update, turn completion, and successful Blynk alarm-event transmission.

8. Known Limitations and Design Notes

- The physical sensor is DHT11 rather than the suggested DHT22/SHT31. Its lower accuracy and relatively frequent transient invalid readings make it less suitable for a real hatch; the software therefore uses immediate fail-safe output shutdown plus a short persistent-invalid grace before systemFault is latched.
- The physical fogger/humidifier was not integrated in the final bench prototype. The relay control path, automatic humidity logic, lockdown target change, and manual command were verified.
- The TEC1-12706 was used only for short low-voltage heater functionality tests. A correctly rated DC supply, heatsink/heat spreader, and independent thermal cutoff are still required before sustained real heating.
- The day counter is based on millis(); therefore, incubation timing restarts after ESP32 power loss. A DS3231 RTC or persistent timekeeping solution is recommended for a real 21-day incubation cycle.
- NFR6 requires 24 h local operation without Wi-Fi. A short disconnect/reconnect test passed, but a full 24-hour no-Wi-Fi soak test was not completed.
- NFR7 mutual exclusion between motor activity and display/I2C activity is not explicitly implemented with a motor/display mutex. The current stepper driver does not use the OLED I2C bus, and turns complete well under 10 s.
- One interrupt-watchdog reset was observed in an earlier integration run. It did not recur in later short validation runs, but a longer stability soak is recommended.
- The mechanical tray linkage still requires final physical calibration so the 512-step motor motion produces the desired tray tilt rather than relying on motor shaft rotation alone.
- The Blynk authentication token and Wi-Fi credentials must be replaced with placeholders in any source code shared publicly; the private demonstration copy can retain local credentials.

9. Conclusion

The Smart Egg Incubator uses an ESP32 with multiple FreeRTOS tasks for sensing, control, safety, egg turning, display, alarms, day counting, and Blynk communication. The system can continue operating locally without Wi-Fi, while Blynk provides monitoring and manual control.Testing confirmed automatic and manual actuator control, egg turning, Day-18 lockdown, sensor-fault protection, two-minute alarms, Blynk alerts, and Wi-Fi reconnection. Main limitations are the DHT11 sensor, missing physical humidifier, non-persistent day counting, incomplete TEC safety/power hardware, tray calibration, and lack of long-duration testing.

THE END😊